

EXPERIMENT: Write a Prolog Program for forward Chaining. Incorporate required queries.

Aim

To write a Prolog program to implement Forward Chaining and incorporate the required queries to derive new facts from existing facts and rules.

Objectives

1. To understand the concept of Forward Chaining.
2. To represent knowledge using facts and rules in Prolog.
3. To derive new facts from known facts using logical rules.
4. To execute queries and verify the derived conclusions.

Algorithm

1. Start the program.
2. Define the initial facts in the knowledge base.
3. Define rules that derive new facts from existing facts.
4. Check the available facts.
5. Apply the rules whose conditions are satisfied.
6. Add the newly derived facts to the knowledge base.
7. Continue applying rules until no new facts can be derived.
8. Execute the required queries.
9. Display whether the queried fact is true or false.
10. Stop the program.

CODE

```
facts = {  
    "sunny",  
    "warm",  
    "humidity_high"  
}
```

```

rules = [
    ("humidity_high", "warm", "rain"),
    ("sunny", "warm", "good_weather"),
    ("rain", "stay_home")
]

print("=====")
print("    FORWARD CHAINING")
print("=====")

print("\nInitial Facts:")
for fact in facts:
    print("-", fact)

changed = True
while changed:
    changed = False
    for conditions, conclusion in rules:
        if all(condition in facts for condition in conditions):
            if conclusion not in facts:
                facts.add(conclusion)
                print("\nNew fact derived:", conclusion)
                changed = True

print("\nFinal Facts:")
for fact in facts:
    print("-", fact)

query = input("\nEnter your query: ").lower()
if query in facts:
    print("Query:", query)
    print("Result: TRUE")

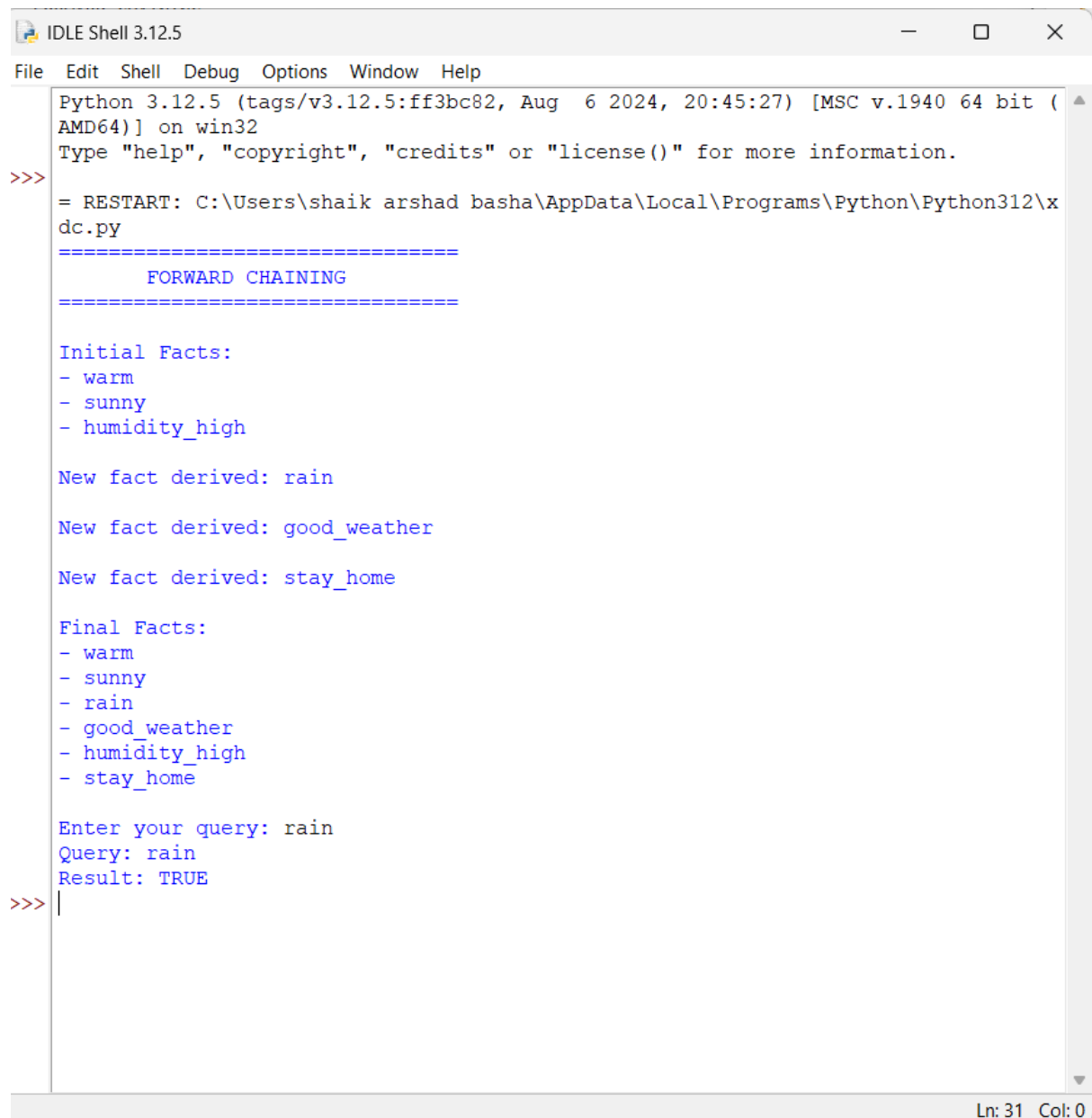
```

else:

```
print("Query:", query)
```

```
print("Result: FALSE")
```

OUTPUT



```
IDLE Shell 3.12.5
File Edit Shell Debug Options Window Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\shaik arshad basha\AppData\Local\Programs\Python\Python312\python.exe
dc.py
=====
          FORWARD CHAINING
=====

Initial Facts:
- warm
- sunny
- humidity_high

New fact derived: rain

New fact derived: good_weather

New fact derived: stay_home

Final Facts:
- warm
- sunny
- rain
- good_weather
- humidity_high
- stay_home

Enter your query: rain
Query: rain
Result: TRUE
>>> |
```

Result

The Forward Chaining algorithm was successfully implemented in Prolog. New facts were derived from the existing facts and rules, and the required queries were successfully executed.

Conclusion

Forward Chaining is a data-driven reasoning technique in which the system starts with known facts and repeatedly applies rules to derive new facts. The Prolog program demonstrates how facts can be progressively generated until the required conclusions are reached.